UNILAK RWAMAGANA CAMPUS

SOFTWARE ENGINEERING AND USER CENTRED DESIGN

Citizen Issue Reporting and Escalation System

CIRES

Milestone 2

Software Requirements Specification

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What we plan to build

We plan to build CIRES for residents who report road or drainage issues, the officers who handle them and the supervisor who assigns and follows up cases. We chose one pilot area so we can test the whole journey from reporting to resolution. The concerns PP1 to PP5 come from our Milestone 1 walkthroughs and still need confirmation through user interviews.

Our pilot covers submission, assignment, tracking, one overdue escalation and resolution with an action summary. Emergency dispatch, SMS, district-wide rollout, external integrations and escalation to several levels are outside this release. We assume that accounts, staff roles and the pilot boundary will be set up before testing.

A reference identifies one report. The resident can use it to find their own case after signing in. Received means the report has been saved. Assigned means it has an officer and a deadline. In progress means the officer has started work. Resolved means the officer recorded an action summary and closed the case. An escalation is a flag on an overdue open report, so it leaves the work status in place.

From a user problem to a test

PP1 is the difficulty of following up without a reference. FR-01 requires CIRES to save a valid report and give the resident a unique reference. In US-01, the resident asks for proof that the report was received. In T01, we submit a valid report and check that one Received record and its reference appear.

What the system must do

ID	Source and priority	Requirement
FR-01	PP1 · Must	Accept a category, location and description. Save a Received report and show its unique reference.
FR-02	PP2 · Must	Let the supervisor assign a Received report to a pilot officer with a future deadline.

ID	Source and priority	Requirement
FR-03	PP3 · Must	Show the report owner its current status and dated updates intended for the resident.
FR-04	PP2 · Must	Let the assigned officer change an Assigned report to In progress.
FR-05	PP4 · Must	Flag each overdue Assigned or In progress report once and show it in the supervisor’s overdue queue.
FR-06	PP5 · Must	Require a nonblank action summary before the assigned officer resolves an In progress report.
FR-07	PP2 · Should	Let the supervisor filter the report list by category and status together.
FR-08	PP5 · Could	Export weekly counts of resolved reports by category without resident identifiers.

How we will measure quality

ID	Acceptance target
NFR-01	Core actions require sign-in. An unrelated account receives no report data and cannot change the report.
NFR-02	Resident views and CSV exports contain zero private contact fields or internal staff notes.
NFR-03	At least 48 of 50 detail requests finish within two seconds with 1,000 stored cases.
NFR-04	Core forms work by keyboard, show visible labels and have at least 4.5:1 contrast for normal text.
NFR-05	Two first-time residents each submit a valid report within two minutes without assistance.

ID	Acceptance target
NFR-06	At least 99% of scheduled five-minute checks succeed during seven days of pilot observation.

We will test access through both the screen and direct requests. We will inspect page responses and the exported file for private fields. We will record the test conditions and treat these figures as targets until the tests are run.

User stories and acceptance checks

US-01 · Must · 3 pts

As a resident, I want to submit a road or drainage report so I can receive proof that it was received.

Given a signed-in resident enters valid details, when they submit, then one Received report is saved and its reference appears.

US-02 · Must · 3 pts

As a supervisor, I want to assign a report with a due date so I can make ownership clear.

Given an unassigned Received report exists, when the supervisor selects an officer and future deadline, then the officer sees it in their queue and the report becomes Assigned.

US-03 · Must · 2 pts

As a resident, I want to check my report so I can see its progress without calling.

Given a resident owns an assigned report, when they open its reference, then they see its current status and public history.

US-04 · Must · 2 pts

As an assigned officer, I want to start work on my case so I can show that action is underway.

Given a report is assigned to the officer, when they mark it In progress, then the new status and timestamp appear in its history.

US-05 · Must · 3 pts

As a supervisor, I want to see overdue cases in one queue so I can follow up before they are forgotten.

Given an open assigned report has passed its deadline, when the overdue check runs twice, then exactly one escalation is recorded and visible to the supervisor.

US-06 · Must · 3 pts

As an assigned officer, I want to close a case with an action summary so I can explain the outcome to the resident.

Given an In progress report belongs to the officer, when they save a nonblank action summary and resolve it, then the report becomes Resolved and the resident can read the summary.

US-07 · Should · 5 pts

As a supervisor, I want to filter the officer queue by category and status so I can find cases for follow-up.

Given the queue contains mixed categories and statuses, when the supervisor applies both filters, then only matching reports are shown.

US-08 · Could · 5 pts

As a supervisor, I want to export a weekly closure summary so I can review the pilot's workload.

Given resolved cases exist within and outside the selected week, when the supervisor exports that week, then the CSV contains only counts by category for that week.

What we will build first

We put US-01 to US-06 in Must. Submission, assignment and resident tracking take eight estimated points in Session 4. Officer progress, escalation and resolution take another eight in Session 5. Filtering is Should at five points and weekly export is Could at five points. The estimates total 26 points. We will use the same priorities on the backlog board.

Who uses CIRES and what they do

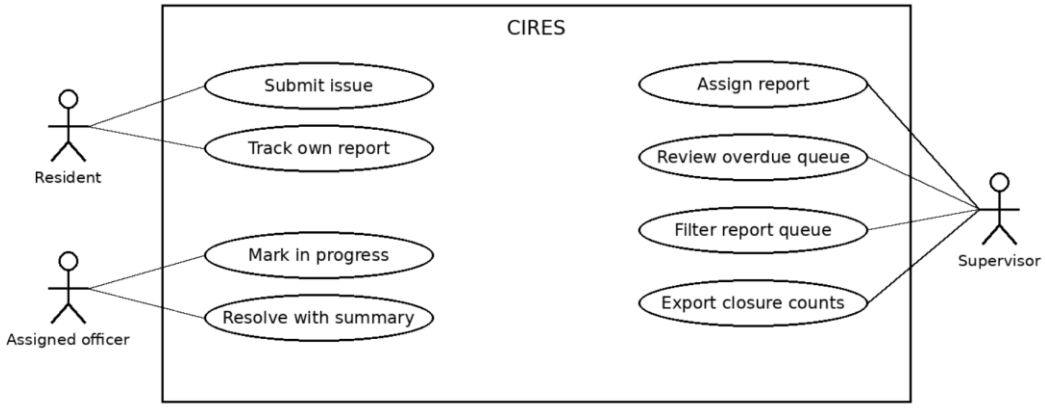


Figure 1. CIRES use cases

UC-01 Submit an issue

The resident signs in, chooses Road or Drainage, enters the location and description, and submits the report. CIRES saves the report as Received and shows its reference. If required information is missing or the location is outside the pilot area, CIRES explains the error and saves no report. This use case covers FR-01.

UC-02 Review an overdue case

An Assigned or In progress report has a deadline that has passed. The scheduled check records one escalation. The supervisor opens the case from the overdue queue and reads its officer, due time and history. Another check keeps the first escalation and timestamp. A resolved case leaves the overdue queue, and its earlier escalation remains in the history. This use case draws on FR-02, FR-05 and FR-06.

How each concern links to a test

Concern	Requirement	Story	Test
PP1 Receipt	FR-01	US-01	T01 Receive a unique reference.
PP2 Ownership	FR-02	US-02	T02 Assign with a future deadline.
PP3 Progress	FR-03	US-03	T03 Show status to the owner.
PP2 Ownership	FR-04	US-04	T04 Record the start of work.

Concern	Requirement	Story	Test
PP4 Delay	FR-05	US-05	T05 Record one escalation after repeated checks.
PP5 Closure	FR-06	US-06	T06 Resolve with a visible summary.
PP2 Ownership	FR-07	US-07	T07 Apply both queue filters.
PP5 Closure	FR-08	US-08	T08 Export counts for the selected week.

Backlog and user review

We will show the Must list to a resident and an officer. We will ask the resident to report a blocked drain and find its status. We will ask the officer to open an assigned road report and explain what they need to act on it. We will record what each person found confusing, what they would change and which requirement we updated. Both reviews are pending. We will add the live Jira or Trello board link and screenshot after setting up the board.